

BOGUCHAR, Russian FederationWMO: **343360**Lat: **49.933N**Lon: **40.567E**Elev: **84**StdP: **100.33**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-19.6	-26.1	0.4	-22.6	-22.8	0.5	-19.2	7.1	-2.3	6.2	-2.3	1.4	50	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	34.4	20.3	32.3	19.9	30.5	19.2	21.9	30.5	21.1	29.6	20.3	28.2	2.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.2	24.6	18.4	13.4	23.9	17.5	12.7	23.5	64.2	30.2	61.3	29.6	58.6	28.3	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.5	4.9	-26.3	36.5	4.4	2.1	-29.4	38.1	-32.0	39.3	-34.5	40.5	-37.7	42.1		
(5)			-26.5	24.3	4.4	2.0	-29.6	25.7	-32.2	26.8	-34.6	27.9	-37.8	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-5.8	-5.0	0.9	9.9	16.5	20.5	22.6	21.4	15.3	8.3	1.7	-3.5		
	DBStd	11.38	6.65	6.76	4.75	4.37	4.63	3.92	3.40	3.88	4.28	4.71	5.25	6.17		
	HDD10.0	2019	491	421	283	55	6	0	0	0	7	89	250	417		
	HDD18.3	3926	749	654	541	254	93	22	4	14	109	311	498	676		
	CDD10.0	1523	0	0	1	52	206	314	392	354	165	37	2	0		
	CDD18.3	388	0	0	0	1	34	87	137	111	18	1	0	0		
	CDH23.3	4229	0	0	0	16	385	908	1454	1254	204	7	0	0		
	CDH26.7	1692	0	0	0	1	117	348	629	547	49	1	0	0		
(14) Wind	WSAvg	2.1	2.2	2.3	2.5	2.3	2.1	1.9	1.8	1.7	1.8	2.0	2.2	2.4		
(15) Precipitation	PrecAvg	514	45	34	30	32	45	59	55	39	41	36	47	50		
	PrecMax	717	104	78	119	109	132	163	180	120	160	123	113	113		
	PrecMin	322	5	1	2	2	4	11	5	1	1	3	0	6		
	PrecStd	111	26	18	20	23	34	37	33	26	35	25	27	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	8.0	15.9	25.4	32.2	34.6	37.6	37.5	30.7	23.6	14.8	8.7		
		MCWB	5.8	6.1	9.0	14.4	19.1	19.9	21.6	21.2	17.6	15.4	11.9	7.5		
	2%	DB	4.4	5.3	12.5	22.5	29.8	32.4	34.6	34.4	27.9	20.1	12.4	6.8		
		MCWB	3.5	4.0	7.1	13.3	18.2	19.8	20.8	20.0	16.8	13.9	10.4	5.8		
	5%	DB	2.8	3.5	9.7	20.2	27.6	30.5	32.3	31.9	25.7	17.6	10.4	5.1		
		MCWB	2.0	2.5	6.1	12.0	17.1	19.2	20.3	19.5	16.3	13.1	8.8	4.2		
	10%	DB	1.7	2.3	7.2	18.0	25.2	28.6	30.1	29.7	23.4	15.3	8.4	3.4		
		MCWB	1.0	1.5	4.5	11.1	16.3	18.7	19.6	19.0	15.4	12.1	7.0	2.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	6.6	10.1	15.8	20.9	22.7	24.0	22.5	20.0	17.2	12.3	7.7		
		MCDB	6.7	7.7	13.7	23.3	30.2	30.5	32.8	32.6	26.1	21.5	13.9	8.3		
	2%	WB	3.7	4.0	8.0	14.1	19.2	21.5	22.0	21.3	18.4	15.2	10.6	5.9		
		MCDB	4.2	4.9	11.7	20.2	27.3	29.0	30.5	31.0	24.6	18.2	12.3	6.8		
	5%	WB	2.1	2.6	6.2	13.0	18.1	20.4	21.1	20.5	17.3	13.6	9.0	4.3		
		MCDB	2.7	3.4	9.2	18.8	25.4	27.7	29.7	29.5	23.3	16.8	10.1	5.0		
	10%	WB	1.1	1.6	4.8	11.8	17.0	19.4	20.4	19.6	16.1	12.2	7.1	2.7		
		MCDB	1.6	2.2	7.1	17.0	23.8	26.6	28.4	27.7	21.8	15.1	8.2	3.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.5	7.6	10.9	11.9	11.6	11.7	13.0	11.7	8.6	5.6	4.9		
		MCDBR	4.0	5.4	11.5	15.0	14.8	14.3	14.9	16.0	15.5	12.7	7.8	4.5		
	5% WB	MCWBR	3.7	4.5	7.2	7.6	6.4	5.6	5.4	5.8	7.0	7.3	5.8	4.3		
		MCWBR	3.9	4.9	10.1	12.5	13.0	12.3	12.6	13.8	12.9	10.4	6.8	4.4		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.355	0.382	0.448	0.413	0.404	0.431	0.438	0.401	0.371	0.347	0.322		
	taud		2.108	2.020	2.095	2.072	2.271	2.327	2.255	2.217	2.292	2.324	2.279	2.184		
	Ebn at Noon		679	747	806	793	846	856	823	796	786	740	658	622		
	Edh at Noon		83	117	130	149	128	122	129	128	107	87	72	68		
(44) All-Sky Solar Radiation	RadAvg		1.04	2.02	3.26	4.37	5.83	6.23	6.07	5.46	3.73	2.04	0.93	0.61		
	RadStd		0.21	0.37	0.51	0.45	0.51	0.52	0.40	0.38	0.43	0.30	0.19	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.62	N/A	N/A	N/A	N/A	N/A	-104	-175	N/A	N/A
(47) Regional (6 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	-149	N/A	N/A

Nomenclature: See separate page